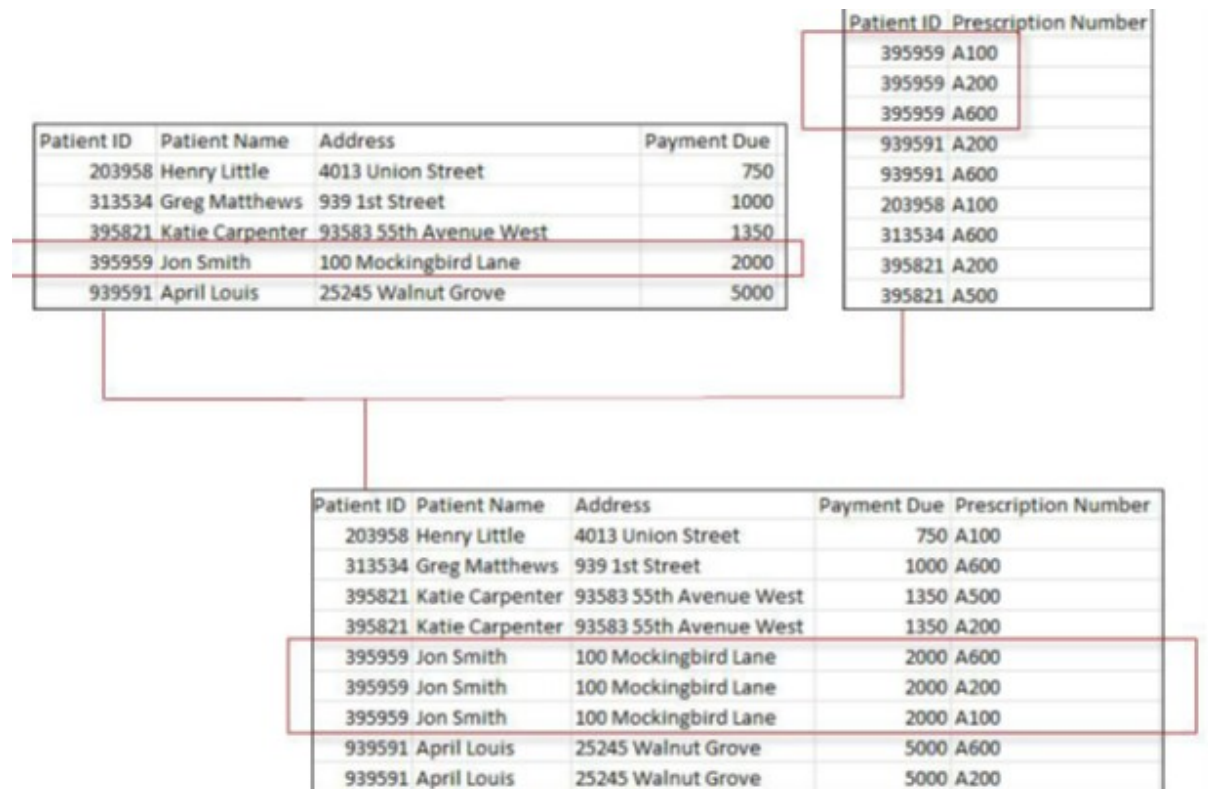


Duplication of Records in Tableau

Cardinality may affect the results returned when joining tables because joining tables that have **one-to-many** or **many-to-many** relationships can have the effect of duplicating values or multiplying rows by the number of times a primary key's match appears in the secondary table.

In the diagram below you can see the left table has a one-to-many relationship with the right table because there are multiple prescriptions for the same patient ID. When these tables are joined, the result table shows that for every time Patient ID 395959 is matched with a prescription in the right table, the values from the left table duplicated the same number of times. Although the payment due for Patient 395959 should be \$2000, aggregating the values in the result table produces the incorrect amount of \$6000 (\$2000 x 3 or the number of times the patient ID appears in the right table).



Solution:-

Tableau offers several solutions for addressing duplication of records when joining tables that have a one-to-many or many-to-many relationship.

- 1) Changing the aggregation from SUM to AVG, MIN, or COUNTD may sometimes resolve the issue.
- 2) Other times, creating a calculated field, using a table calculation or choosing to blend the data will provide a solution.
- 3) Custom SQL is another way to remove duplicates.

The technique you use is dependent upon the data as well as what you are trying to accomplish.

Nested Table Calculations behaviour

A nested table calc is when one table calc is used inside another table calc, which is different from writing a calculation that uses two table calculation functions independently. This is an important distinction because a nested table calc will allow you to compute the parent table calc separately from the child table calc, whereas a calculation containing two independent table calculation functions will act as a single table calc when determining how to compute the table calc,

In the Below Example, we are having two table calculations

1)Lookup 2)Index

While addressing we need to set addressing Options individually for both the functions.

The screenshot displays a Tableau worksheet with a table calculation. The data is organized by Gender (Female and Male) and Quarter of Bet date (Q1, Q2, Q3, Q4). The columns are Daily Number of Bets and Nested. Red boxes highlight specific values in the 'Nested' column, and red arrows indicate the direction of the calculation (downwards).

Table Calculation [Nested]

Calculated Field: **Nested**

Calculation Definition: **LOOKUP (SUM ([Daily Number of Bets]), [Index])**

Compute using: **Table (Down)**

At the level: **Table (Down)**

Restarting every: **Table (Down)**

Description: Results are computed along Gender, Quarter of Bet date.

Formula: LOOKUP(SUM([Daily Number of Bets]),[Index])

Buttons: OK, Cancel, Apply

Table Calculation Functions

1.WINDOW,RUNNING FUNCTIONS:-

WINDOW functions perform aggregations on a selected range of data that is included in the structure of the view. The range is set by negative integers moving backward from a point and positive integers moving forward from a point. The arguments FIRST() and LAST() as well as parameters and other fields can also be used to include all data from a given point to the beginning or end of the data set.

Gender	Year of Bet date	Daily Number of Bets	Running Sum
Female	2006	501,014	501,014
	2007	1,163,612	1,664,626
	2008	1,631,737	3,296,363
	2009	1,060,207	4,356,570
	2010	395,031	4,751,601
Male	2006	3,009,565	3,009,565
	2007	5,387,630	8,397,195
	2008	9,853,831	18,251,026
	2009	14,010,902	32,261,928
	2010	5,417,814	37,679,742

Running Sum

Results are computed along Pane (Down).

```

RUNNING_SUM(SUM([Daily Number of Bets]))

```

The calculation is valid.

Sheets Affected ▾

Gender	Year of Bet date	Daily Number of Bets	Window Sum
Female	2006	501,014	4,751,601
	2007	1,163,612	4,751,601
	2008	1,631,737	4,751,601
	2009	1,060,207	4,751,601
	2010	395,031	4,751,601
Male	2006	3,009,565	37,679,742
	2007	5,387,630	37,679,742
	2008	9,853,831	37,679,742
	2009	14,010,902	37,679,742
	2010	5,417,814	37,679,742

Window Sum

Results are computed along Pane (Down).

```

WINDOW_SUM(SUM([Daily Number of Bets]))

```

The calculation is valid.

Sheets Affected ▾

Gender	Year of Bet date	Daily Number of Bets	Window Sum
Female	2006	501,014	
	2007	1,163,612	501,014
	2008	1,631,737	1,664,626
	2009	1,060,207	3,296,363
	2010	395,031	4,356,570
Male	2006	3,009,565	
	2007	5,387,630	3,009,565
	2008	9,853,831	8,397,195
	2009	14,010,902	18,251,026
	2010	5,417,814	32,261,928

Window Sum

Results are computed along Pane (Down).

```

WINDOW_SUM(SUM([Daily Number of Bets]), FIRST(), -1)

```

// All integers {..., -2, -1, 0, 1, 2,...}
// Any integer-type field, calculated field,
// table calc, or parameter
// FIRST(), LAST()

The calculation is valid.

Sheets Affected ▾

2.LOOKUP:-

LOOKUP looks forward (positive integers) or backward (negative integers) a specified number of dimension values in the view. Backward and forward can be determined either by the structure of the view (i.e., left to right or top to bottom) or by a specified measure to sort

Gender	Year of Bet date	Daily Number of Bets	Lookup
Female	2006	501,014	
	2007	1,163,612	501,014
	2008	1,631,737	1,163,612
	2009	1,060,207	1,631,737
Male	2006	3,009,565	
	2007	5,387,630	3,009,565
	2008	9,853,831	5,387,630
	2009	14,010,902	9,853,831
	2010	5,417,814	14,010,902

Lookup

Results are computed along Pane (Down).

`LOOKUP(SUM([Daily Number of Bets]),-1)`

// Any integer {..., -2, -1, 0, 1, 2,...}

The calculation is valid.

Default Table Calculation

Sheets Affected ▾

Gender	Year of Bet date	Daily Number of Bets	Lookup
Female	2006	501,014	
	2007	1,163,612	
	2008	1,631,737	
	2009	1,060,207	501,014
Male	2006	3,009,565	
	2007	5,387,630	
	2008	9,853,831	
	2009	14,010,902	3,009,565
	2010	5,417,814	5,387,630

Lookup

Results are computed along Pane (Down).

`LOOKUP(SUM([Daily Number of Bets]),[Years back])`

// Any integer-type field or calculated field
// or table calc or parameter

The calculation is valid.

Default Table Calculation

Sheets Affected ▾

Years back

-3
-1
-2
-3
-4

Gender	Year of Bet date	Daily Number of Bets	Lookup
Female	2006	501,014	
	2007	1,163,612	
	2008	1,631,737	
	2009	1,060,207	501,014
Male	2006	3,009,565	
	2007	5,387,630	
	2008	9,853,831	
	2009	14,010,902	3,009,565
	2010	5,417,814	5,387,630

Lookup

Results are computed along Pane (Down).

`LOOKUP(SUM([Daily Number of Bets]),[Years back])`

// Any integer-type field or calculated field
// or table calc or parameter

The calculation is valid.

Default Table Calculation

Sheets Affected ▾

Years back

-3
-1
-2
-3
-4

3.INDEX&SIZE:-

INDEX counts 1, 2, 3,... up through however many values there are in a given dimension or combination of dimensions.

SIZE tells you the number of values in a given dimension or combination of dimension. You can think of SIZE as the maximum INDEX value.

Gender	Year of Bet date	Daily Number of Bets	Index
Female	2006	501,014	1
	2007	1,163,612	2
	2008	1,631,737	3
	2009	1,060,207	4
	2010	395,031	5
Male	2006	3,009,565	1
	2007	5,387,630	2
	2008	9,853,831	3
	2009	14,010,902	4

Index

Results are computed along Table (Across).

INDEX ()

Default Table Calculation

The calculation is valid.
Sheets Affected ▾
Apply
OK

Gender	Year of Bet date	Daily Number of Bets	Size
Female	2006	501,014	5
	2007	1,163,612	5
	2008	1,631,737	5
	2009	1,060,207	5
	2010	395,031	5
Male	2006	3,009,565	4
	2007	5,387,630	4
	2008	9,853,831	4
	2009	14,010,902	4

Size

Results are computed along Table (Across).

SIZE ()

Default Table Calculation

The calculation is valid.
Apply
OK